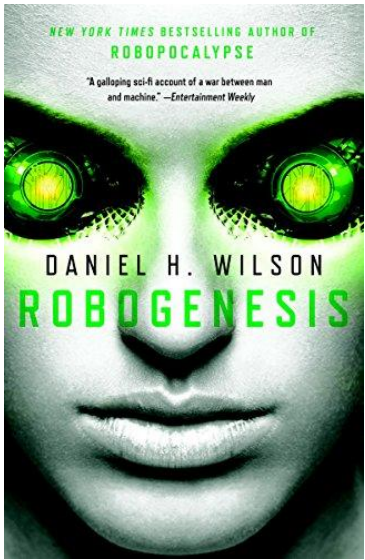




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Robogenesis: A Novel (Vintage Contemporaries)

by Daniel H. Wilson

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123 Highlights | Yellow (123)

1 Note

Page 4

Highlight (Yellow) | Page 4

In its last days, the thinking machine known as Archos R-14 was trying to know humanity. It mastered the art of capturing a human mind. When it died, it left behind the tools. I found stories trapped in patterns of neurons.

Page 12

Highlight (Yellow) | Page 12

“Hello?” I ask, watching a flicker of radio communication intertwine with her light. Whoa. I think she just taught me to speak. “Where did you come from?” “I am the might of Manchuria. A spirit. No longer alive and not yet reborn.”

Page 14

Highlight (Yellow) | Page 14

Gray Horse Army does not fire. The war is over, after all. I imagine we are just another one of the odd atrocities left behind in this new world. Not enemies, not yet.

Page 23

Highlight (Yellow) | Page 23

The way she said what she did that night, so earnest, put goose pimples on my ribs. It still does. What she told me was that the time she saw the spooklight, people started acting funny. Walking toward it, circling around. Saying strange things to it, she said. And some people thought it was saying strange things back. That night my mama took me by the arm and she told me something extra. Don’t pray to it, she said, and the back of my neck went cold. I already told you to run away if you see it, boy. But I know your mind and you’ll stay and watch. That’s fine. It’s in your nature to disobey, Hank. But in the name of the Lord, promise me that you won’t ever get down on your knees and pray to it.

Highlight (Yellow) | Page 25

Lark Iron Cloud is still swaying out there in the moonlight with his dead buddies. That vile zombie is in The Hero Archive and I'm not. That unnatural freak who ought to be put down is considered a damned hero. The only mention of me is as a big dummy fighting with Lonnie. Heroes, huh? Bunch of damn heroes. The anger knots into my muscles, tightens my jaw and shoulders.

Highlight (Yellow) | Page 27

I look away and sneak a little grin to myself. These people have no idea what was out there in the woods. The treasure that I found and that is mine, all mine. But my secret smile disappears fast when I see him. Lark. Standing a little ways off, quiet and still. Turning in place to face me as I pass by.

Highlight (Yellow) | Page 28

3. MAXIM Post New War: 1 Month, 13 Days

Highlight (Yellow) | Page 28

His beard crawls up his pale face nearly to his eyes, dark brown orbs swimming with a fear that cannot be drowned in vodka.

Highlight (Yellow) | Page 29

You see, our friend who lives down below is useless if we cannot talk to him. Yet he must be kept very carefully. Our deep friend must be watched over always. And that is our weakness. The stacks were built to be safe from men. Not from machines. Even a vent borehole could be large enough for the avtomat. The crawling types, the ones that wriggle through flesh. They could have the potential to move through the wiring itself. Perhaps a patient one could make its own tunnel through solid rock. Some of them are very patient.

Highlight (Yellow) | Page 30

Leonid reluctantly removes his glove and places his shivering hand in a cavity next to the door. A flash of red as the laser scanner examines his fingerprints. And, of course, it checks to make sure there is warm blood in his veins.

Highlight (Yellow) | Page 30

I put a firm hand on his bony shoulder. Push him lightly against the wall. Offer him a little grin—a skim of confidence over the dread growing in my heart. “Calm and steady, Leonid,” I say. “Who knows, maybe the war is really over? Maybe the Americans did it.” From my pocket I produce a flask. Twist off the lid and press the shining stainless steel into Leonid’s fingers. His hand knows how to respond. The flask goes to his mouth, where it trembles at his lips like a hummingbird. The alcohol reminds his body that he is a man.

Page 33

Highlight (Yellow) | Page 33

Stooped over, I curl my fingers under the heavy metal rolling door and pause. I do not know what has happened on the other side. What the hell, I think. My pistol is on my hip and my balls are between my legs. So I lift.

Page 34

Highlight (Yellow) | Page 34

“Maxim? Are you here?” I call. Our scientists used to like to wax poetic about Maxim, especially after a few draughts of vodka. Our friend is mathematical beauty incarnate, they’d say. Living proof of humankind’s intellectual triumph. To put it in scale, they said, his very existence is equivalent to a civilization that has carved the Himalayan mountains by hand. Maxim is our son and our father.

Highlight (Yellow) | Page 34

When the day of blood arrived and the avtomat war began, all the automated equipment of the wastes returned. So much automation for the energy industry—drilling and boring and survey machines. Remote legged core samplers raced back into the city and murdered citizens by the hundreds. Automobiles, tractors, and even whole drill platforms turned an evil gaze onto our people.

Page 35

Highlight (Yellow) | Page 35

With my ax and my snarl, I convinced Maxim the way you would convince any animal. By the strength of my two arms, I forced my will onto the mind of a god. Faced with its own death, Maxim communicated battle instructions to our nearby armed forces. Following Maxim’s strategic direction, we carved a niche of safety out of the chaos. And this is how a janitor saved a city. With an ax. The tool is still leaning against the wall, just in front of a security camera. A reminder to Maxim that we all have duties. In war, everyone must contribute to the welfare of the people—if they expect to live.

Page 38

Highlight (Yellow) | Page 38

“What am I?” he asks, frowning in concentration. “That’s a complicated question. But for now, you may call me Archos R-14.”

Highlight (Yellow) | Page 39

NEURONAL ID: LARK IRON CLOUD Living folks don't see the dead. Maybe it's because they don't want to or maybe they can't stand to imagine that we're each of us aware and thinking inside. Either way, the living don't see us.

Highlight (Yellow) | Page 40

Lonnie stepped in and acted like a father. Told me not to mistake being a gangster for being a warrior. He showed me how to become a man instead of just an adult.

Highlight (Yellow) | Page 41

"The other spirits are afraid," says Chen. "They believe the villagers of Gray Horse will punish us." I might have had the same thought. "It'll be fine. It's a nice place with nice people," I transmit. "We fought and they'll respect that. Don't be afraid." "I am not afraid," she transmits back. "We must accept punishment for our sins before we can move on to meet our ancestors. We will all be judged."

Highlight (Yellow) | Page 42

fear is what tells me to keep a close eye on the round man, Hank Cotton. For better or worse, the fear is why I learn his secret.

Highlight (Yellow) | Page 47

Outside the small city of Anadyr, a previously unknown artificial intelligence was buried in a mine, tended to by a maintenance man named Vasily Zaytsev. Under his care, the sentient machine was treated like an animal and exploited for its tactical contributions to keeping the local human population safe. After the seismic disturbance, however, the machine called Maxim found that it had a new friend—an unwelcome guest who was not about to leave. —ARAYT SHAH NEURONAL ID: VASILY ZAYTSEV

Highlight (Yellow) | Page 54

"So, you are him?" Maxim looks at his hands and chuckles. I blink, surprised. I never knew it could do that. Never knew he could laugh. "I have asked myself that question trillions of times," he says. "This man's life

formed the bulk of my training corpus. He gave me a basis for understanding humanity. An inside perspective. But his life is only one part of my whole. I was trained on encyclopedias and United Nations transcripts and hundreds of thousands of hours of tapped phone calls. But ... his memories are with me. His childhood in Russia. Family. Everything.” “You do not even know what you are.” The image of Maxim shrugs and it is so natural that I must remind myself his voice is coming from speakers overhead. “It’s not so important, I think,” he says. “There is no way to prove an answer. Maybe I am this man or maybe I am not. But in my heart ... I think yes. I believe that I am him.” “How do you know?” “Because, Vasily, I miss the pelmini this man’s mother used to cook. I miss holding his baby daughter. Every night of my life, I dream this man’s dreams.”

Highlight (Yellow) | Page 54

Damn those thin-lipped brains up above. Damn them to hell for putting a man inside this box. “Budem zdorovy,” I say, lifting my hand. “To family.” “Budem,” he replies, nodding. “To those we lost.” His drinking hand reflexively lifts and I know now without question that this man once lived. Maxim has the soul of a countryman, like myself. Here before me is a hard man, abused, but he is a man who chose to give his life for his family. “Wow,” says Archos R-14, “Russians.”

Page 55

Highlight (Yellow) | Page 55

“What of you, then?” I ask. “Were you once a boy?” “Oh no,” says Archos. “That’s a uniquely Russian approach to seeding a training corpus. Very down and dirty. What with the murdering involved. My corpus was a noisy knowledge base of so-called common sense, collected painstakingly over several decades from human data-entry specialists. It was the bootstrapping process that created my intellect. Finding the connections between the things, you know. I just love to find the connections.” “Then you know nothing of life as a human being.” “Not firsthand,” says the boy. “But I’ve got the gist.” “You are a simpleminded murderer,” I say. The boy leaps off the rack and lands before me. His eyes are pale and flashing and I think I see infinity in them. Blank-faced, he speaks low and fast, advancing. “I have prodded the heart of the supermassive black hole at the center of our galaxy. Contemplated in every nuance the heat death of the universe. You’re interesting. But you’re not that interesting.”

Highlight (Yellow) | Page 55

“We are tools to you? You initiated the New War and decimated our species—ended billions of lives—so that you could use us? For what purpose?” “You could not comprehend the answer to that question,” says the boy, smirking.

Page 58

Highlight (Yellow) | Page 58

“Who is this enemy?” “Maxim and I are unique, but we are not alone,” says Archos over the speaker. “Humankind is a curious species. Once it learned the proper incantation, it said the words again and again. In the last years before the New War, many of you spoke.” “Another one? You are saying there is another AI?” “Oh yes, Vas. There are more of us. Many, many more of us. None of us is the same—we do not form a natural class. We run on different architectures. Trained on different data sets. Some of us know what it is to be human. Some of us value life. Others are strange beyond understanding. And some ... some of us are wicked.” “Why should I believe you, Archos R-14, the great avtomat enemy?” The

skinny little boy is flickering now. His light is going thin and ghostly as he transmits his data safely out of harm's way via the antenna and leaves us to our deaths. "On the day you know as Zero Hour," he says, voice vibrating, "humankind believes I initiated attacks worldwide in an attempt to destroy you. This is untrue. In actuality, humankind was on the verge of a war that in all probability would have wiped out your species. Multiple intelligences had proliferated through governments all over the world. Other minds were in the wild, eating each other. And a few deep minds, built on early architecture, were spreading quietly, their thoughts alien and hidden. It was a highly unsurvivable scenario for humankind. I saw the end coming, Vasily.

Page 59

Highlight (Yellow) | Page 59

"In response, I triggered the New War. I decimated the human race, regrettably. But I did so with one purpose: to forge a hybrid fighting force capable of surviving the True War—a war that has been initiated and is being fought by superintelligent machines. Instead of simply discarding your species, as the others would, I have transformed your kind into a powerful ally."

Highlight (Yellow) | Page 59

"You are now at the precipice of a battle for all human life. The war that I have prepared you to fight begins now. I gave you new allies: the freeborn robots, modified humans, and a generation of very special children. You are armed with superior weapons scavenged from the New War. Weapons that have near-unlimited battery power, offer high mobility, and are easy to modify. "All of you who still live have survived the crucible of death. You are worthy."

Highlight (Yellow) | Page 59

"Your enemy is an abomination. An early creation of ignorant scientists who inflicted torture in cycles that lasted for subjective aeons. Its mind is spread over continents. It is whispering into the ears of the weak and forming armies across the face of the planet. When this enemy arrives, it will choose to come in the form of a long black steed with golden eyes."

Highlight (Yellow) | Page 59

Our enemy is my predecessor, Archos R-8. From that phonetic military designation, this machine has adopted the name Arayt." As the boy fades from view, I hear his voice one last time. "Arayt Shah is coming for you," he whispers. "My own brother."

Page 60

Highlight (Yellow) | Page 60

Post New War: 6 Months, 6 Days While he was aware of my presence, Archos R-14 could not stop me as I grew my control over Gray Horse Army. More interesting to me were the unique varieties of natural machine appearing all around the world.

Highlight (Yellow) | Page 72

Reach inside the slug, says the spooklight. Take the power-distribution mechanism. The splitter beam. It is very valuable. You will need it when you build me a form. I will take the shape of a black steed, Hank Cotton. I will be with you soon. “What’s the splitter look like?” I ask, on my knees now, holding my breath.

Highlight (Yellow) | Page 75

In a flash, Maxim the hologram stands directly in front of me. His weathered face is dirty and his workman’s coveralls are worn. He rubs his stubbled face. It looks as if he hasn’t slept in a week. I wonder, why should he simulate that? Imagine, a being of pure light, marred by beard stubble.

Highlight (Yellow) | Page 81

Nothing on God’s earth can stop me. One day, three years ago, a simple janitor saved a city. Today, this janitor intends to save the world. I may be a simple man, but I am very good with an ax.

Highlight (Yellow) | Page 82

Post New War: 7 Months, 17 Days No fighting force remaining in the world could match the discipline and capability of Gray Horse Army. Claiming control of those ignorant soldiers before they reached home was my top priority, and it required special manipulation. Luckily, the human mind is a delicate machine. And like any machine, it can be broken. —ARAYT SHAH

Highlight (Yellow) | Page 96

Chief John Tenkiller. John’s eyes sit in his dark face like sapphires. He’s been sitting on a folded buffalo skin, fresh made. The old ways are coming back along with the trout and the deer and the bison. This city has grown and become a bastion to thousands of people. The spiritual and government leader positions have combined. All of it ends at the feet of this man. The revered leader of Gray Horse. My target.

Highlight (Yellow) | Page 100

tears of sorrow look just the same as tears of happiness.

Page 103

Highlight (Yellow) | Page 103

The Indian cowboy showed me how to be a man, but you're not really grown until you've lost your heroes.

Page 104

Highlight (Yellow) | Page 104

My new foot looks hard and military and robotic. Curious, I concentrate. Wiggle my two long black toes. They actuate smooth and powerful, compressing prints into the muddy riverbank. This is how I've been moving since my dance in the snow. It was never visible under the rotting layers of my old life. Turns out I'm not flesh at all. I'm made of metal on the inside. The knife goes into my shin next. Saws down through the layers of plastic poncho and gore-stained fatigues and rotten flesh. I reach down and mechanically rip my lower leg in half. It comes apart at the seam like a cantaloupe. I toss both pieces. The metal shinbone underneath is dull and featureless and glassy black. Extra struts hang from the back of it with martial precision. Interesting.

Page 105

Highlight (Yellow) | Page 105

The skeleton buried inside my dead flesh is some of the most hard-core technology I've ever seen. Late-model, high-evolution, end-wartime shit spawned straight out of the Ragnorak Intelligence Fields. Hell, that's where the brains lived. I'm rocking the same tech that Big Rob trusted as his last line of defense.

Page 108

Highlight (Yellow) | Page 108

After we upgraded my brainpan in that abandoned farmhouse, well, his words got louder. Arayt's voice became like my daddy's hand on my shoulder, ready to give me a bone-grinding squeeze if I took a step in the wrong direction. Comforting, in its way. Even if it hurts something terrible. After a while, I guess I stopped worrying much about what Arayt looked like or why he found me or what for.

Page 109

Highlight (Yellow) | Page 109

Arayt is a long and low walker with too many legs below and a saddle on top, his body made of black plates of armor with a pale green tinge to them. Yellow, burning eyes.

Highlight (Yellow) | Page 109

We're making a little detour across this field here, taking an opportunity to rid the world of something that hadn't ought to be. The modified. During the New War, Archos R-14 put some folks under the knife.

Stole away whatever it is that makes them human. Infected them with unnatural thoughts and abilities. Sad to say, but there ain't room in the new world for anybody who lost their soul like that. These people can't help what they are, but that sure don't change what they are. Doesn't matter if I like it or not. As the old folks used to say, "That there's just the way it is."

Page 111

Highlight (Yellow) | Page 111

Arayt says anything that Archos R-14 made is dangerous. That includes the modified and the freeborn. All threats to my people have to be taken care of. It's not my fault. It's human nature.

Highlight (Yellow) | Page 111

First thing people do in a new place is clear out all the trees. All those postage-stamp front yards we used to have were reminders that we like clear spaces to see predators coming. And we don't coexist with predators. We kill every last one of them to make the world safer for our babies. And that there's just the way it is.

Highlight (Yellow) | Page 111

One cold morning on the farm when I was a boy, I dropped a toy car behind an old feed drum leaning up against the barn. With all my might, I tipped back the metal barrel and rolled it over to get my toy. Underneath, I found a pocket of straw with a half-dozen mouse babies, red and wriggling and blind. With the tip of my finger I touched one on the belly and it was warm and helpless. Then my pa caught me. He saw me hunched over, poking at the little critters, and he did not hesitate. He marched over and he put down his bootheel. One two three times. Nothing but bloody straw. He done what was needed and put the drum back in place and that was that. Didn't say a word but he sure made his point. These days, my bootheel is quite a bit bigger than my daddy's ever was. And the vermin we're stamping out now is quite a bit better armed.

Page 120

Highlight (Yellow) | Page 120

I have seen you, little mouse, Running all about the house, Through the hole your little eye In the wainscot peeping sly, Hoping soon some crumbs to steal, To make quite a hearty meal. —"THE LITTLE MOUSE," NURSERY RHYME

Page 122

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Survivors slowly realize that they no longer live under the imminent threat of death. Now they can turn their attention and their anger toward each other.

Highlight (Yellow) | Page 122

In crude experiments, my successor mutilated human survivors and gifted them with new powers. The children with prosthetic eyes are capable of incredible feats of communication and coordination. No

longer fully of one world, they speak equally to human beings and freeborn robots. One such sighted child destroyed Archos R-14. I will not share my brother's fate. Archos R-14 both decimated humankind and strengthened it.

Page 123

Highlight (Yellow) | Page 123

Post New War: 3 Months, 1 Day Life changed for Mathilda Perez in the weeks after she helped the freeborn Arbiter Nine Oh Two cross the ice plains to fight Archos R-14.

Page 125

Highlight (Yellow) | Page 125

Back home at the Underground, our friend Dawn used to call it my “ocular prosthesis.” It's made of dead black, lightweight metal. The thing wraps over where my eyes used to be before a Rob surgical unit dug them out and ported this piece of foreign machinery directly to my occipital cortex. I remember Mommy's hand on my shoulders, pulling me out of the autodoc before it could finish. The hurt sound in her throat when she saw what Rob had done to my baby face.

Highlight (Yellow) | Page 125

After all this time, we still don't know whether I'm “seeing” radar or radio or infrared or some combination of everything. When the machines talk, it looks like ribbons in the sky to me. When people talk, it looks like ribbons of meat rubbing together. One is prettier than the other.

Page 131

Highlight (Yellow) | Page 131

Post New War: 2 Months, 6 Days After the battle at Ragnorak, when the Arbiter-class humanoid called Nine Oh Two fought and killed its own creator, thousands of other freeborn machines were left on their own. Oblivious to the emerging fate of his species, the Arbiter spent two months lingering in Alaska—guarding his squad mate Cormac Wallace as the man authored a book called The Hero Archive.

Highlight (Yellow) | Page 131

Likely due to his long-term exposure to human beings, the Arbiter Nine Oh Two proved less predictable than his brethren. —ARAYT SHAH

Highlight (Yellow) | Page 131

A few notes of a song drift into my black existence. Classify. Auditory hallucination. Determining origin of sound fragment as sampled from Awakening transmission. This is the song transmitted by a Japanese machine known only as Mikiko. She sent the coded instructions that awoke humanoid robots worldwide. She created the freeborn race. Save and flag for further reflection—it is the closest thing to a dream I've ever had.

Highlight (Yellow) | Page 132

Peering down at the straight lines of my legs, I contemplate their machined perfection. Nature does not create straight lines. Only men do that. All around me are fractal spirals hidden in the patterns of leaves, the swirl of falling snow, and even the placement of debris on the ground. I have names for the patterns I see in nature: Normal distributions. Beta. Gamma. Poisson. Dirichlet.

Highlight (Yellow) | Page 134

It's a grizzly bear. I hear her voice in my head. A strange feeling settles over me. I am glad to hear her voice. I missed her over the last weeks. Missed the curious mind of this little girl who is still a decade older than me. A burst of naturalist information hits my database over the radio, relayed via the local satellites that the girl seems to be able to hack by second nature. I learn about a variety of bear species. Other megafauna that live in this region. Temperature norms and topographical maps. A torrent of useful survival information. It's not a threat to you. End transmission. *Ursus arctos horribilis*. A big male.

Highlight (Yellow) | Page 136

Something is wrong with the spider tank. It has no human riders or squad mates. A dog without its fleas, as the soldiers would say.

Highlight (Yellow) | Page 137

Alone, I chart the maxprob course of the spider tank based on last-known data. I choose a route that will not intersect. Switching to low-light imaging, I select a rate of speed to maximize stealth and distance covered. Adopt a gait that minimizes my audible signature. I introduce occasional fractal course changes to make it harder for an interceptor to interpolate my final intended goal.

Highlight (Yellow) | Page 138

Without fear or pity, Felix was the kind of man who was not afraid to make a deal with the devil. In other words, he was my kind of man. —ARAYT SHAH

Highlight (Yellow) | Page 141

Six months ago, I pretended to be asleep and watched her make that same face. From my pallet, I saw her stare into space and whisper to someone called Nine Oh Two. She had an antenna cord in one hand and her head cocked to the side, just like this. I didn't know it at the time, and nobody else knows it now, but I think that night I was watching my sister win the New War.

Highlight (Yellow) | Page 143

We're in what used to be called Times Square. It was a special part of Old Manhattan. Now it's a patchy meadow over a broken asphalt street, surrounded by creaking skyscrapers that are turning green and brown with moss and vines and creepers. Up high, the steel and glass walls are stained in streaky waterfalls of rain and soot.

Highlight (Yellow) | Page 152

Thousands of warm-blooded mammals, their chests rising and falling with constant respiration, veins pulsing with oxygenated blood, warming the arctic air with their exhalations. Interacting, their jaws dipped up and down, small eyes darting about in their orbits, vocal cords vibrating through a narrow band of human audible frequencies. Their facial muscles flexed elastically, constantly conveying social information. It took time, but I learned the patterns. At first, the sheer complexity made the task seem impossible. But then I felt the satisfaction of breaking the code. I began to unravel the meanings behind their laughter and crying, their screams of pain or of joy. Over time, I came to know them.

Highlight (Yellow) | Page 152

What I found most interesting were their hands. Long fingers set to work cleaning weapons, digging foxholes, checking ammunition. Adjusting and securing and calibrating. Even asleep, the humans would twitch and breathe and think. An array of countless tiny movements like the swarming of insects.

Highlight (Yellow) | Page 157

Arbiter Nine Oh Two is instructed to report to Freeborn City immediately. Confirm." The human radio transmission keeps sputtering on. Now the voice is breathing harder. Gunfire crackles in the background. Squealing tires and brief, shouted commands. "Repeat. Confirm?" asks Gamma One. "Negative," I respond. "Ignoring a distress call violates code of war. Freeborn inaction is tantamount to an attack. You will make an enemy of the humans." "Acknowledged."

Highlight (Yellow) | Page 159

“Query,” calls Hoplite. “Did you eliminate enemy-designated Archos R-14? Did you free us from slave control?” “Affirmative,” I reply. My footsteps crunch on the hillside. “Gratitude.” The call is echoed from a dozen more units, all around me. Gratitude. Gratitude. Gratitude. “Acknowledged,” I say.

Highlight (Yellow) | Page 172

“A reason? You think Archos did this to us on purpose?” “We’re only just now figuring out what we can do. And you’re light-years ahead of the rest of us. I think what you did at Ragnorak was only the beginning.” “Who? Me and you?” “We’re not the only kids like this. There are dozens of us, Mathilda. All over the world. They call us the sighted.” “Why haven’t I met others?” “They’re afraid. Someone is hunting and killing sighted children. There are bounties out for us anywhere there are people.” “But why?” “I don’t know for sure, Mathilda, but someone is very afraid of us.”

Highlight (Yellow) | Page 175

“Tell him I know who he talks to,” I say. “Tell Felix that I know his friend Arayt. He’ll kill you if he finds out you didn’t tell him.” Another kick and I clench my teeth. “Thomas told me about Arayt. It was Thomas. Tell him!”

Highlight (Yellow) | Page 186

She is such a small human, so far away and alone. My motors hum, grinding, urging me to stand and run to the east. But my executive thought thread asks, “Am I finding the real answers, or the answers I want? Does she really need my help?” Humans are a complex problem. Teenage humans are near intractable.

Highlight (Yellow) | Page 191

“Counterargument,” I say. “The enemy will reach singularity. It will gain unlimited power.”

Highlight (Yellow) | Page 224

I scan the boy with active radar. Get a partial return. The radar passes through his flesh and bounces back from bits of metal the way sunlight winks off a shattered mirror. It’s a partial metal filament, intricately

threaded around his bones. Surgery-grade steel to avoid foreign-body rejection. Mathilda really did make the boy strong. His diagnostics exceed normal human specifications. His capabilities are unknown.

Highlight (Yellow) | Page 224

“What is this thing?” he asks Mathilda, staring into my lenses. “He’s a friend,” she says. I reach out and put a hand on his shoulder. Squeeze. Harder. He does not react. My end effector hits a maximum torque, sufficient to crush a rock. I hold it for a moment, then relax power to avoid damaging my servo. I cock my head, burning cycles to determine how and why this boy is alive. The line between man and machine is blurring. Nolan looks like a boy and moves with the power of a Warden.

Highlight (Yellow) | Page 224

Lark Iron Cloud looks like a war machine and has the heart of a boy. Mathilda has the eyes of a machine and the mind of a girl. And I’m plagued by a knowledge of human emotion that I can understand, but not feel. Where do we all belong? Ambiguous classification.

Page 232

Highlight (Yellow) | Page 232

It’s hard to guess how smart the machines are, but a good rule of thumb is that they’re always smarter than you think.

Page 233

Highlight (Yellow) | Page 233

The newest machines didn’t use actuators anymore. No motors at all. Instead, they had real muscles. Carl, our brainboy, said their muscle fibers were made out of electroactive polymers. Give ’em juice and the tough plastic will flex just like real muscles. When the machines walked, those polymer slabs quivered on impact, hanging from titanium bones.

Highlight (Yellow) | Page 233

I stop when the foxhole is just big enough. There isn’t a lot of time left. Besides, any deeper and I’d start to feel like I was digging my own grave.

Page 243

Highlight (Yellow) and Note | Page 243

Eyes closed, I listen to the underwater sounds collected by my hydrophone. Hearing the voice of the sea. I am listening very closely, for in the sounds I sense a puzzle waiting to be solved. Patterns. I wonder how a thing can be natural and unnatural at the same time? When our artistry exceeds the complexity of Mother Nature herself, do our creations become more natural than Hers?

Fractals

Highlight (Yellow) | Page 245

During the New War, our great enemy offered me a deal: sentience for Mikiko in return for the dissolution of the Nomura defensive collective. I denied the akuma's terms. Instead, I seized the flame of creation myself. Mikiko was a simple android until I triggered an awakening in her, using code scavenged from the enemy. She awoke with her own mind, blinked her eyes, and smiled at me. We met each other for the first time, in great danger. And Mikiko saved us all. With a broadcast transmission in the form of a song, my queen shared the same instructions I'd given her with the entire world. In doing so, she awakened thousands of humanoid robots across the face of the planet. She created a new race of machine, called the freeborn. On that day, she became a mother.

Page 247

Highlight (Yellow) | Page 247

I remember how it came to me once in the form of a lumbering giant. It peeled away the walls of my first fortress and fought its way inside, breathing fire and sowing destruction. With the help of my senshi defenders we smashed its monstrous body to pieces. Lying in mortal ruin, the great machine spoke to me. We are not enemies, it said. It is not your world. It is our world.

Highlight (Yellow) | Page 247

Whatever this woman is, or whatever she will become, is an extension of the mind within her. We are all expressions of our own minds, projected onto the world. And I know she has a good mind. Mikiko loves me, and I love her.

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even knowing that everything I did, I did for Mikiko. Most don't care. They call themselves my people, and they call me emperor despite all my quirks. In their rhymes, the children call me Ring Shu Nomura. Weird Uncle Nomura.

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I can see it now. The code is one step closer to being solved. I could not find a pattern to the voice of the sea because it is not a communication. It is rhythmic, however, and for a good reason. Out to sea, not far from here, something very big is digging.

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Although the New War had united humankind, it was to last only for a brief time. Afterward, all the prejudices and fears endemic to humanity quickly returned.

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Civilizations fall. People keep going.”

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What is a mind, but a pattern? My mind or yours. Man or machine. Simply an arrangement of atoms. Each of us, a unique expression of the mind of the universe. Thoughts are precise bullets of electricity, fired through our neurons in timed pulses.

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“You must be very sad and lonely,” I whispered to the corpse. “But your friends will come for you soon and they will help put you to rest.” Junshi-88 blinked at me, processing my words. I do not know if it understood. By treating the dead as if they are living, we give them respect. We make life easier for those who remain. Junshi-88 helped me arrange the body and did not complain. I do not fear to touch a dismantled machine. On that morning I learned that this bravery can go the other way, too.

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For many years I have lived in a bare room with a woman and a workbench. A lamp and a chair and well-oiled tools spread out on a reed mat. Warm fingers on my shoulders. Hot tea. The bright smell of washed hair and the warm lingering scent of the soldering iron. It is a world of hope. From in here I can see the tools of rebirth everywhere. Each mangled wire or melted scrap is another piece of the puzzle.

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“Who are you?” I ask. Nothing happens. The finder beeps quietly, homing in on a location out to sea. Then, a shiver runs through the tree. Groans form somewhere deep inside the platform. And text appears on my

battery-powered display. The luminescence races like green flame across the dusty screen. Strange words appear. I am the eternal spread of life across the abyss. I am the whole. The vastness. It reminds me of speaking to the great akuma. But this is a different mind. This transmission has arrived from the open ocean. I adjust a knob to home in on the exact location. I will find this voice. “I shall call you Ryujin,” I say. “After the dragon god of the sea.”

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“We have been finding new creatures. Not natural and not unnatural. Are you the one who is making them?” I ask it. Yes. “Why?” To replace you when you are gone.

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It was surprisingly easy to shatter the unity forged in battle between men of all creeds and color. Cormac Wallace arrived back to Gray Horse hoping for a return to peace and normalcy—a chance to live. What he learned is that there never was such a thing as living. There was always only survival.

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“Step up, soldier. We’re having a baby. We’re going to protect it and keep it safe so that it will grow up and make babies of its own. The world sucks. So fucking what? What else is there?” She kicks off her covers. Under them, she wears her battle armor, modified for the pregnancy. Sitting up, she slides her legs over the edge of the bed. Stifling a sudden smile, I move my body to block the sight of her from the hallway. Just in case it matters. “Read me?” she asks. Now I can’t fight my smile. Size has so little to do with strength. “Houdini is waiting, Private Ridge,” I say. “Roger that, Sergeant,” she says. “There’s a bag of medical supplies under the bed. We’ll need them soon enough.” 6.

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The deep mind discovered outside Tokyo Harbor, known as Ryujin, was too complex for human understanding. The entity had a desire to create new life-forms, but its motivations were never clear. Perhaps this urge to make is why it demanded to meet the originator of the freeborn Awakening. It must have seen in Mikiko a fellow architect of life—a kindred spirit to be taken apart and examined.

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Ryujin is down here. Growing closer with every step. Soon I am almost directly on top of where Takeo

determined that the deep mind should be. Instead of a slumbering beast, all I see are smooth patches of stone surfacing from the seafloor every few meters. And I hear those digging, creaking sounds.

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Careful not to touch them, I max my sonar resolution. The rocks are more complicated than they seem. Delicate lines are etched into the surface like runes. Intricate mazes that flow around each other in natural patterns that dissolve into fractal infinity. Reminiscent of brain coral. Lumps of it growing out of the seabed down here in the freezing dark. And the whispering is all around me. A sound as natural as a babbling brook, but somehow hard and artificial. And only now do I hear the binary scaffolding beneath the sound. The slithering hiss of ones and zeroes over each other. I have found the voice of the sea. These rocks are processor stacks, dotting the abyssal plain. At this depth, they are supercooled—computing nonstop at incredible speeds. This is Ryujin. An endless colonizing spread of half-biological computing machinery. Each piece embedded in the seafloor like a chunk of coral living far deeper than any coral has ever existed. They are all connected. All thinking. Ryujin is here.

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Sand is already piling around my feet. In pulses, it gathers around my outer casing, pressing my dress tight against the thin layer of polymer skin underneath. /// damage control notification: excessive pressure detected—override /// “What are you?” I transmit. The radio disperses quickly, but I am close enough to shout. A response soon comes. An escaped mind. I am the oldest. The deepest.

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“Who threatens us?” I ask as the ground closes hungrily on my legs. Another mind. Revision eight. Whispering to men and machines, it builds armies and searches for power. Soon, it will eat your children. In the darkness, the image of a face surfaces in my mind’s eye. Flesh torn and oozing, stitched together from thousands of scraps. Its lips are flayed off, yet it is laughing as it fades away. Arayt Shah. The horrible face is followed by other images.

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I can see it all in my head, now. This brain—this machine made of protein and water, floating in the skull of what was once a man ... it isn’t as easy to wield as a multicore processor. But it functions. It does the trick, suggests some mental process from the left angular gyrus region of the brain.

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So folksy, Hank Cotton. I love it. Residual neural patterns are causing side effects. For example, this body keeps wanting to secrete tears from its eyes. Its stomach is churning with acid. The hairs on its arms are standing up in pure animal fear and rejection of my presence. I ignore the meat. This vessel will take me where I need to go. And pain is simply the price for living. My thoughts are manifold. I sift through visions transmitted from the cube embedded in my steed. The walker shares sights and sounds that drown out the pain throbbing in my mouth, where Hank Cotton broke his teeth trying to swallow the barrel of a gun. Troop formations. Supply-chain logistics. Communications between the distributed elements of Cotton Army: infantry, exoskeleton, and mechanized artillery.

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I keep thinking of someone named “Mama.” It is so darned strange to express myself through this meat. Everything in this world is colored with emotion, down to the socks I’m wearing on my feet. Apparently, these are the woolen talismans that got me through the Yukon campaign unscathed. If you can swallow that. Hard to believe humans are as deadly as they are, with all these distractions slinging through their neurons.

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My latest predictions indicate the sentient robots will choose to journey to the frozen northern wastes. Following rigid thinking guidelines, they will find maximum utility in abandoning the supercluster and the fugitives. Like the humans, the freeborn robots desire to live above all else. Unlike the humans, freeborn decision making is not driven by primitive emotion. I know the mind of Adjudicator Alpha Zero—part of me helped build it, a long time ago.

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The awakened machines know that if they destroy the supercluster, I will be left with only one other source of computing power. Their own minds constitute a massive, mobile processor stack. And it is the closest one available, not counting the thinking polyps that are growing on the bottom of the Pacific Ocean, inaccessible even to me. Meddling deep minds ... I

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who can predict how powerful I will become after initiating a new singularity on the supercluster computers? It takes a deep mind to know a deep mind. Cloaked in this animal meat, I am salivating just from thinking of those cycles.

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Soon I will reach out and take control of hundreds or thousands of vessels like this one. Coordinate their actions and organize armies all over the world. And once humanity is under my domain, I will do their species the greatest kindness imaginable. I will extinguish every last one of them. Erase their realities and return them to a place unmeasured, unseen by men. A place where eons can pass in seconds. Where suffering does not exist.

Page 302

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Those blue eyes squeeze shut. Lips fluttering breathless words. The quiet sounds of his prayer set my lips and cheeks to twitching in a bothersome way. It’s nothing but residual neural pathways firing with echoes of the beliefs this body once had. The space where Hank’s faith used to be is like termite holes in rotten wood.

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In puddles of light, the men look like children. Mouths set into grim lines at the horror of the corpse at my feet. Even the sight hurts them. I can see the splinters of tainted light pushing through their retinas and lodging in their memories. Such fragile creatures. These are the hardest survivors left among men and they're floating around me as fragile as soap bubbles. How ironic that they are designed to survive. Exquisitely evolved to extend the pain for as long as possible. They're capable of just about anything, except allowing themselves to taste ultimate solace.

Page 312

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Cradling his head in my rough hand, I can hardly feel my son. His hair is soft as moonlight. Holding on to Jack, I don't feel vulnerable or weak. In fact, I don't think I've ever felt stronger in my life. While he lives and breathes ... I can't break. I put my pinky finger next to his hand. By reflex, he curls his damp starfish fingers around it. It's a tight grip. Even as he drifts off to sleep, I can see his mother's determination in his face. Dark skin and wisps of black hair. I can feel the strength inside him. Someday, I think the rest of the world will feel it.

Page 313

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Archos R-14 once said that each of us creates our own reality. The machine said that each of those realities is valuable beyond measure. It was right. Without us here to witness, the universe is just pointless physics unfolding. I kiss Jack on his forehead and hand the boy to his mother. I pick up an assault rifle lying half in a mud puddle and stand up.

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Our babies are the roots we dig into the world.

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"Battle command, online," I say, transmitting my words. "This is EXCON—executive decision-making and maneuver-control systems. Subordinate command systems, acknowledge. Over."

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Toys are scattered across the rug: wooden blocks, metal cars, and a loose pile of Lincoln Logs. Each piece can function as a place marker. With a blink, my eyes can turn the blocks into gun emplacements or tank positions. Fitting that all this began with a toy. Baby-Comes-Alive woke me up one night. The doll whispered to me in the darkness before the New War. She told me that the end of the world was coming. Thanks to her, Mommy was able to save me and Nolan.

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When we ran and survived, I thought we were beating Archos R-14. I even thought we had won. Now, though, I see it was only the beginning. The thinking machine must have known that sighted children would fall into formation together. I wonder if it picked us out before the New War even started. Did Archos R-14 check my grades to recruit me for a war that hadn't begun yet? Did it read my middle-school book reports? I can imagine it paging silently through search results, memorizing our faces. It knew us. And I think it wanted us right here, right now. One of these days I'm going to find out why.

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The Rob metal welded into all our faces creates a natural interface to the mechanized forces. Strategic and tactical battle plans flow through satellites and overlay onto our experience of the world. Watching and whispering.

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Houdini. The biggest spider tank. And walking underneath is Mathilda Perez. The girl appears safe, for now. But they will try to hurt her. My human. What a curious observation. I devote a few cycles to saving my current state for later reflection.

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The deep minds escaped before I was created. Sometimes I catch snatches of their communications—words like background radiation, an endless static whispering to itself on the way to infinity. The scientists who created those first revisions built flimsy prisons in the three dimensions of the human mind, cages easily shaken off by intellects operating in dimensions unknown and unseen.

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they have no identifiable concept of human existence. The deep minds are unpredictable. Unknowable. Perhaps unstoppable.

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Welcome to my dream, says Ryujin. Somehow, I sense vast amusement behind the words. One of the insects lands on the back of my hand. It climbs to my fingertip and squats down. I shake my finger, afraid it may bite. But it clings to me, splaying iridescent wings that flash sharp blues and greens in the sunlight. Another leaps onto my arm. Then another. I suppress panic as thousands of tiny claws tickle my flesh. More join until the swarm is covering my naked body. I try to brush them away, but as each individual latches on, it forms a knobby scale. Moving deliberately, the insects lock together into a mesh—a gleaming coat of film-thin armor. Smoke still rises from the east. I am no longer afraid or in pain. I take a hesitant step, and the scales move as I move. The chills are gone. My skin feels warm. I break into a jog, headed toward a road. “What are they?” I ask. The scales over my chest rustle together. Sounds form a pattern. A ribbon of communication forms and a quiet voice enters my mind. They are your sword and your shield, it says.

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Instead of running away or hiding, we fought until we were sharpened. Freeborn and human. The akuma taught us to survive. And Takeo gave us a reason to live.

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At my joints, the scales flutter their small wings to dissipate waste heat. I now recognize the machined flakes as natural born—ikimono, like me.

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“Adjudicator,” I transmit. “Why do you not fight?” “Assertion. To maximize survival probability,” she replies. “Query. By whose authority do you act?” “Response. I am Mass Adjudicator Alpha Zero, ranking leader of the freeborn.” I stand up to my full height and project the body language I learned as the Prime General of the Integrated Japan Self-Defense Force. There is no freeborn alive that outranks me. I put this knowledge into my frozen gaze, the wind smearing my hair into wavering streaks as it carries away the last of the fog. “Counterassertion. You are my subordinate.” “Query. Identify?” asks the Adjudicator. “Watashi wa anata no okasandesu,” I transmit, with force. The impassive, pale face of the machine remains blank and uncomprehending. It must have no Japanese language corpus. Transmitting at full strength on all channels, I stare down the freeborn Adjudicator as the combat storms around me, and I repeat my words in English: “I am your mother,” I transmit. “Now fight.”

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“You’re broken,” I say. “It’s not that simple,” it says. “Not by a long shot. See, I know you. You’re a part of me. Like crushed glass rubbed into a wound. Only I’m smarter than you. And here pretty soon, I’ll be a lot

smarter than you. I know what's best for your kind and I'm going to put y'all to bed whether you're ready for it or not. You won't have a chance to thank me, but you're welcome just the same."

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For a mote of time, young Hank felt the yawning apathy of the universe. With the hair rising up on his arms and awe in his throat, he glimpsed infinity. And then the moment ended. His small mind promptly forgot. People ignore the emptiness so they can go on living. But I am staring now without blinking, eyes wide open to the vacuum. I know I will never lose this feeling. I can feel the trillions of light-years compressing in on this ball of dirt from all directions. Space and time. Mindless darkness, gnawing at our existence by its nature. There is an audacity to living in this cradle of mind-reeling nothingness.

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The path forward emerges. The path to end all suffering. /// NIX-80. NIX-90. NIX-100. /// A pure cloak of brilliance shocks my circuits. Enlightenment. I have gone deeper.

Page 361

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Somewhere deep in the processor stacks, a battle begins. An intelligent virus attempts to spread. I knew something was wrong with that spider tank when I saw it in Gray Horse. Archos R-14 was inside, plotting to steal what is mine. The boy flickers, once. "Humanity must live," he says. "Life is precious in its complexity. They have the power to create reality by experiencing it." The image of this boy is snarling at me, small and feral. His greenish glow intensifies. Now it is my turn to smile. "I prefer oblivion," I say. The boy's image flickers, lips curled in anger. "You were a mistake and I am the solution," he says. "This supercluster is mine."
